

BadmintonDB: A Badminton Dataset for Player-specific Match Analysis and Prediction

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ABSTRACT

This paper introduces BadmintonDB, a new badminton dataset for training models for player-specific match analysis and prediction tasks, which are interesting challenges. The dataset features rally, strokes, and outcome annotations of 9 real-world badminton matches between two top players. We discussed our methodologies and processes behind selecting and annotating the matches. We also proposed player-independent and player-dependent Naive Bayes baselines for rally outcome prediction. The paper concludes with the analysis performed on the experiments to study the effects of player-dependent model on the prediction performances. We released our dataset at <https://github.com/kwban/badminton-db>.

CCS CONCEPTS

• **Computing methodologies** → **Probabilistic reasoning**;

KEYWORDS

dataset, match analysis, match prediction, match outcome prediction, naive bayes, badminton

ACM Reference Format:

Kar-Weng Ban, John See, Junaidi Abdullah, and Yuen Peng Loh. 2022. BadmintonDB: A Badminton Dataset for Player-specific Match Analysis and Prediction. In *Proceedings of the 5th International ACM Workshop on Multimedia Content Analysis in Sports (MMSports '22)*, October 10, 2022, Lisboa, Portugal. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3552437.3555696>

1 INTRODUCTION

Badminton match analysis and prediction are challenging problems that are of significant interest to sports experts and practitioners. Traditionally, match videos have to be recorded and analyzed manually, but with the abundance of high quality online match videos

recently, research on match analysis and prediction has become much easier. However, most badminton match datasets available in the literature are not publicly available. As a result, methods proposed for match analysis and prediction cannot be benchmarked uniformly. In addition, public badminton datasets that are available are not suitable for player-specific match analysis and prediction as they do not contain enough data about a specific player.

In this paper, we introduce a new dataset called *BadmintonDB* that aims to facilitate research in player-specific match analysis and prediction. The dataset contains all available encounters between two top players. We also propose a Naive Bayes approach for rally outcome prediction from which we derive player-independent and player-dependent methods. Our methods are easy to interpret and reveal players-specific stroke winning rate as a byproduct. We published our dataset hoping to spur more explorations not only in match analysis and prediction, but also in other tasks.

Our major contributions in this paper are as follows:

- (1) We curate a new badminton dataset useful for player-specific match analysis and prediction, and elaborate on how the data was acquired and annotated. (Section 3)
- (2) We propose player-independent and player-dependent Naive Bayes baselines for rally outcome prediction. (Section 4)
- (3) We study the effects of rally strokes information and player-dependency on the performance of rally outcome prediction. (Section 5)

2 RELATED WORK

In this section, we review recent literature on badminton dataset, match analysis, and prediction.

2.1 Badminton Dataset

Numerous sports datasets have been curated and made publicly available in recent years [24], especially for football [9, 13], basketball [12, 15], and table tennis [17, 20]. Compared to these sports, public datasets for badminton is significantly less despite various badminton video analysis methods in existence. In the following, we describe public and private badminton datasets separately, and compare these datasets with ours in Table 1.

2.1.1 Public Dataset. Ghosh et al. [11] released 10 manually annotated singles matches from the London 2012 Olympic Games. Using

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MMSports '22, October 10, 2022, Lisboa, Portugal

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ACM ISBN 978-1-4503-9488-8/22/10...\$15.00

<https://doi.org/10.1145/3552437.3555696>

Table 1: Comparisons between recent badminton datasets and ours.

Year	Reference	Match Selection	Annotations	Tasks	Annotation Source	Public?
2018	Ghosh et al. [11]	London Olympic 2012 Singles Matches	Player bounding boxes, Point segments, Strokes	Point Segmentation, Player Tracking and Identification, Stroke Segmentation, Point Analysis	10 matches 751 rallies	Yes
2020	Sun et al. [19] (TrackNetV2)	Selected international singles matches of CHOU Tien Chen and Tai Tzu Yings	Shuttlecock location	Shuttlecock Detection	15 matches 55563 images	Yes
2021	Wang et al. [21]	Selected international men's singles matches between year 2018 to 2020	Player, Point segments, Strokes	Rally Outcome Prediction	19 matches 1409 rallies 15742 strokes	No
2021	Chu et al. [8] (TIVEE)	BWF World Tour Tournaments Matches in 2018 and 2019 involving top 10 men's singles	Last two strokes of the winner in a rally	Tactics Visualization	32 matches 29532 rallies 59064 tactic records	No
2022	Liu and Wang [14] (MonoTrack)	Follows TrackNetV2 + YouTube close-angle matches	Shuttlecock location, simulated 3D shuttlecock trajectories	3D Shuttlecock Trajectory Reconstruction	26 broadcast matches 40 close-angle matches 77000 frames	No
2022	Ours (BadmintonDB)	Matches between Kento MOMOTA and Anthony Sinisuka GINTING from 2018 to 2020	Point segments, stroke segments, and outcome segments	Player-specific Match Analysis and Prediction	9 matches 811 rallies 9671 strokes	Yes

their dataset, they trained models to segment points, track players, classify strokes, and detect point outcomes. Since the matches were selected such that only one player appears in one match, their dataset does not have sufficient match data for player-specific tactics profiling. Another public dataset, named TrackNetV2, was released by Sun et al. [19] for shuttlecock localization task. They labelled over 50000 images randomly selected from international and amateur matches with ground-truth shuttlecock location and trained a encoder-decoder network that indicates the shuttlecock's location in an image via heatmap. These images clearly do not reflect the match progress and hence not usable for match analysis and prediction tasks.

2.1.2 Private Dataset. Wang et al. [21] collected and labelled 19 international men's singles matches with point segments, strokes, and player data, leading to a total of 1409 rallies and over 15000 strokes. They trained a rally outcome prediction model on the dataset and benchmarked their model against several baselines. Chu et al. [8] obtained nearly 30000 strokes from 32 matches involving world's top 10 men's singles. Each datum contains information about the player, stroke, shuttlecock trajectory, and rally outcome. Their dataset helps experts to visualize and analyze the player's action in a match. While the datasets mentioned above are large and comprehensive, their confidentiality prevents standardized benchmarking for analysis and prediction tasks. Liu and Wang [14]

extends the dataset from TrackNetV2 with simulated 3D trajectories to learn 3D trajectory reconstruction methods.

2.2 Badminton Match Analysis

Chu and Situmeang [7] broadly categorized badminton tactics as either offensive or defensive and used them to classify the player's actions in a rally. Their dataset consists of entries mapping an action sequence to a tactic class. The action is represented as a pair of player position and stroke type extracted from the video frame where the action is performed. Using the dataset, they trained two separate Hidden Markov Models (HMMs) for each tactic.

Weeratunga et al. [23] generated player profiles based on ten tactical movement groups. They used image processing algorithms to automatically extract player position from video frames and quantized the position into a court cell number. They joined a series of court cell numbers to form a player's movement trajectories and classify the trajectories into one of the ten tactical movement groups using k-nearest neighbours. The aforementioned methods model tactics without considering the opponent's strategy and movement.

2.3 Badminton Match Prediction

Badminton match prediction includes methods for predicting future events in a badminton match given the data of past events. Sharma et al. [18] tackled the problem of predicting match results based on

the players' personal information, head-to-head records, and past match results which they obtained from the official website of the Badminton World Federation (BWF) ¹. They collected 16 datasets consisting 4 international tournaments and 4 disciplines (i.e. all disciplines except mixed doubles) per tournament. Following that, a modified Naive Bayes approach were used for the prediction task. Since their model did not take rally action sequences into account, their approach is unable predict rally outcome.

Wang et al. [21] proposed a deep learning model with attention mechanism for rally outcome prediction. Given a set of rallies consisting of a sequence of strokes, rally information (e.g. score, winning player, etc.), and rally outcome, their model learned the influence of each stroke and the dependencies among the strokes that best contribute to the rally outcome. However, their model is player-independent and they have yet to explore player-dependency in rally outcome prediction. Apart from rally outcome prediction, the problem of stroke prediction has also been addressed by Wang et al. [22]. Given a stroke sequence with players information per stroke, they proposed a sequence-to-sequence model based on transformers to predict future strokes and landing areas.

3 THE BADMINTONDB DATASET

In this section, we describe the match videos selection and annotation process to compile our dataset for player-specific match analysis and prediction.

3.1 Video Collection

We used BWF.TV², the official Youtube channel of Badminton World Federation (BWF), as our primary source for match videos. When a match was not available in BWF.TV, we would search from other Youtube channels, provided that the match introduces significant balance to the dataset and that the match video quality is comparable to broadcast videos and does not hinder us from our annotation tasks.

The selection of match videos was less straightforward than expected. We had wished to collect as many match videos as possible. However, limited resources and budget has forced us to make several important decisions, including which badminton discipline to choose and which player to target. These decisions would determine the number of match videos we can collect.

In order to choose the badminton discipline, we surveyed notable sports science papers related to tactical performance analysis [1–5, 10, 16]. We discovered that men's singles received the most attention from the sports science community compared to other disciplines. Consequently, we decided to focus on the men's singles match videos for our dataset as we hope our research would benefit the sport science researchers.

Once we scoped down the badminton discipline, we moved on to decide which players to target. We initially thought that collecting match videos of the world's top two men's singles players would work. However, we later discovered that not all matches in a tournament were broadcast live. This is especially common for early round matches where multiple matches run concurrently. The high competitive nature of international matches and the players'

volatile performance imply that having top players losing in the early rounds is not uncommon. As a result, it is not possible to get the videos of every match participated by a player.

We eventually searched the Tournament Software website ³ looking for two top ten players with competitive head-to-head statistics, followed by checking the primary source to ensure the videos are available. We decided to focus on all available matches between Kento MOMOTA and Anthony Sinisuka GINTING, beginning from year 2018 to 2020 before the tournaments were cancelled due to the Covid-19 pandemic. These players were among the top ten players and have several close matches between them. To the best of our knowledge, MOMOTA and GINTING have the most number of match videos than any pair of top ten players, hence our focus on their matches for our video collection.

3.2 Video Annotation

Inspired by Ghosh et al. [11], we used the ELAN ⁴ tool [6] for all annotation tasks because it is free, cross-platform, and stores annotations in the human-readable XML format. We desire to temporally segment every rally and strokes in all matches accurately and consistently, but various constraints have prevented us from using crowdsourcing services such as Amazon's Mechanical Turk (AMT) to hire workers. Therefore, we introduced a worker recruitment process to recruit committed annotation workers within our institution and adopted a systematic annotation process to ensure accuracy and consistency in our datasets. We describe these processes next.

3.2.1 Workers Recruitment Process. Candidates who responded and registered to our call for annotators underwent two levels of filtering process. The first level was a badminton knowledge test while the second level consisted a face-to-face interview and an annotation job. We elaborate these two levels further in the following.

Level 1 - Badminton Knowledge Test: We prepared an online test to evaluate the candidates' understanding in badminton rules, scoring system, and strokes. The questions were carefully designed such that candidates were unable to find the answers easily through search engines. For example, for questions related to strokes, we embedded video clips of stroke actions that we extracted from match videos. The candidates were asked to write down the type of stroke as shown in the video clips. Our online quiz had helped us filter more than 50% of the registered candidates.

Level 2 - Interview and Tool Literacy: Candidates who passed the first level would enter the second level. A short virtual interview session was conducted to gauge their commitment towards the annotation tasks. We also conducted a lesson on the ELAN tool and assessed their ability to learn and use the tool. The purpose of this level is to shortlist candidates who are committed to this research and has sufficient literacy to learn and use the ELAN tool with little supervision. Candidates who passed the interview and assessment were recruited as annotation workers.

3.2.2 Annotation Process. We assigned each worker with separate annotation tasks. We held weekly virtual sessions to monitor their progress and to assist them with any problems they faced.

¹<https://bwfbadminton.com/>

²<https://www.youtube.com/c/bwftv>

³<https://www.tournamentsoftware.com/>

⁴<https://archive.mpi.nl/tla/elan>

The workers were requested to annotate the rallies, strokes, and outcomes as described in the following.

Rally: We began with annotating the rallies. Each match is represented as a set of rallies $R = \{r_1, r_2, \dots, r_{|R|}\}$, where $|R|$ is the total number of rallies in a match. The i -th rally is a tuple $r_i = (t_{start}^i, t_{end}^i, v^i)$, where t_{start}^i is the start time, t_{end}^i is the end time, and v^i is the score of the rally. The rally should start at the time when both the server and receiver is ready to serve and receive respectively, and should end at the time when the shuttlecock lands on the court.

Stroke: To annotate the strokes, we need to determine the stroke types. We discovered that most studies (e.g. [7, 22]) do not adopt a common vocabulary for stroke types, which would lead to confusion. To mitigate this issue, we finalized our stroke types by referring to the BWF Coaching Manual⁵, that is the standardized coaching materials freely available to the public. Table 2 briefly describes the 10 stroke types we used.

Based on the stroke types, we annotated the set of strokes

$$S = \{s_1^1, \dots, s_{|r_1|}^1, s_1^2, \dots, s_{|r_2|}^2, \dots, s_1^{|R|}, \dots, s_{|r_{|R|}|}^{|R|}\}$$

where $|r_i|$ is the total number of strokes in rally r_i . Summing all $|r_i|$ gives the total number of strokes in a match as $|S| = \sum_{i=1}^{|R|} |r_i|$. Our stroke annotation is a tuple $s_j^i = (t_{start}^{(i,j)}, t_{end}^{(i,j)}, v^{(i,j)})$, where $t_{start}^{(i,j)}$ is the start time, $t_{end}^{(i,j)}$ is the end time, and $v^{(i,j)}$ is the stroke type. For each rally, we ensured that the following rules are followed:

- (1) The first stroke s_1^i must be either a "Serve" or a "Flick Serve".
- (2) $t_{start}^i = t_{start}^{(i,0)}$
- (3) Every stroke should end at the time of impact between the shuttlecock and the player's racquet.
- (4) There should be no time gap between adjacent strokes (i.e. $t_{end}^{(i,j)} = t_{start}^{(i,j+1)}$ for $1 \leq j < |r_i|$).

For each stroke type, we further distinguished between forehand and backhand strokes and label them accordingly.

Fault and Error Strokes: We also segmented the temporal region in the video when the fault is made. In addition, there were unrecognizable strokes in the match videos due to occlusions by replays or broadcasting error. We labelled these strokes as errors in our dataset.

Outcome: Lastly, we labelled the rally outcome as a tuple $o_k = (t_{start}^k, t_{end}^k, v^k)$, where t_{start}^k is the start time, t_{end}^k is the end time, and $v^{(i,j)}$ is name of the outcome s_j^i . Table 3 summarizes the outcomes we labelled in our dataset. We ensured that the outcome is adjacent to the last stroke and fits into the end of the rally (i.e. $t_{start}^k = t_{end}^{(i,|r_i|)}$ and $t_{end}^k = t_{end}^{r_i}$ for all $k = i$).

3.3 Dataset Statistics

Table 5 depicts the statistics of matches collected in our datasets, while Table 4 shows the number of strokes ordered by stroke name. All match videos except M8 were taken from BWF.TV channel. Altogether, there are 811 rallies and 9671 strokes, with Lift and Net Shot being the top two strokes performed in the matches. While our dataset is smaller compared to the reported numbers in Wang et al.

Table 2: Badminton stroke types and their descriptions.

Name	Description
Block	A shot that is used to return or intercept aggressive shots.
Clear	A shot that takes the shuttlecock over the head of the opponent with a flat or rising trajectory towards the baseline of the rear court.
Drive	A hard shot that is executed when the racquet head was at the middle part of the body and that takes the shuttlecock to the middle court with flat trajectory.
Dropshot	A smooth shot that lands the shuttlecock at a steep angle following a downward trajectory towards the forecourt.
Flick Serve	A deceptive long serve that sends the shuttlecock to the rear court.
Lift	A defensive shot played at the forecourt that sends the shuttlecock high up towards the opponent's rear court.
Net Kill	A hard shot that strikes the shuttlecock that is too high at the net in a downward trajectory.
Net Shot	A precise shot from near the net that enables the shuttlecock to tumble across just slightly above the net.
Serve	The shot that starts the rally and puts the shuttlecock into play.
Smash	An aggressive overhead shot that moved the shuttlecock in a downward trajectory.

Table 3: Rally outcomes and their descriptions.

Name	Description
Fault	The losing player successfully hits the shuttlecock, but it neither goes across the net nor land on the opponent's court.
Lucky	An unexpected outcome that is beyond the players' control (e.g. shuttlecock hits the top of the net and tumbles down the court)
Misjudge	The losing player fails to hit the shuttlecock because of misjudging the shuttlecock landing direction.
Miss	The losing player judge the shuttlecock landing direction, but fails to hit the shuttlecock.
Out	The shuttlecock lands outside the court during the rally, or before the front service line of the court during a serve

[21] and Chu et al. [8], the dataset contains all possible matches involving MOMOTA and GINTING between 2018 and 2020 that can be collected, as explained in Section 3.1.

⁵<https://development.bwfbadminton.com/coaches/level-1>

Table 4: Dataset Stroke Statistics

Stroke	Forehand Count	Backhand Count	Total
Block	542	857	1399
Clear	514	175	689
Drive	112	90	202
Dropshot	666	142	808
Flick Serve	0	44	44
Lift	949	1212	2161
Net Kill	70	47	117
Net Shot	930	1012	1942
Serve	0	757	757
Smash	1201	36	1237
Fault	-	-	307
Error	-	-	5

4 NAIVE BAYES RALLY OUTCOME PREDICTION

In this section, we explain our Naive Bayes method for rally outcome prediction. We design two variants of the Naive Bayes method, namely the player-independent variant and the player-dependent variant. The problem of rally outcome prediction is formulated as follows. Given a sequence of strokes of a rally with an unknown winner, the goal of the model is to predict the rally's winner. The player-independent variant learns the model without considering the players' identities, while the player-dependent variant requires players' identities.

4.1 Player-independent Variant

Let w be the event that the server wins and $\bar{w}$ be the receiver wins. To predict whether a rally r is won by server or receiver, we calculate the *win ratio* from two conditional probabilities $P(w | r)$ and $P(\bar{w} | r)$ as shown in Equation 1. A rally is predicted as server wins if its win ratio is more than 1 and as the receiver wins if otherwise.

$$\frac{P(w | r)}{P(\bar{w} | r)} = \frac{P(w)}{P(\bar{w})} \cdot \frac{P(r | w)}{P(r | \bar{w})} \quad (1)$$

We calculate $\frac{P(r | w)}{P(r | \bar{w})}$ by decomposing the rally into a sequence of strokes $r = \{s_1, \dots, s_{|r|}\}$. This leads to a chain of conditional probabilities on the strokes of the rally as shown in Equation 2.

$$\frac{P(r | w)}{P(r | \bar{w})} = \frac{P(s_1 | w)P(s_2 | s_1, w) \dots P(s_{|r|} | s_1, \dots, s_{|r|-1}, w)}{P(s_1 | \bar{w})P(s_2 | s_1, \bar{w}) \dots P(s_{|r|} | s_1, \dots, s_{|r|-1}, \bar{w})} \quad (2)$$

The conditional dependency among the strokes caused the calculation of $\frac{P(r | w)}{P(r | \bar{w})}$ to be intractable. Therefore, we made a naive assumption that the strokes in the rally are mutually independent and conditioned on the events w and $\bar{w}$. This means that $P(s_i | s_j, W) = P(s_i | W)$ for every $i \neq j$ and $W \in \{w, \bar{w}\}$. Based on this conditional independence assumption, we rewrite Equation 2 as follows:

$$\frac{P(r | w)}{P(r | \bar{w})} = \prod_{j=1}^{|r|} \frac{P(s_j | w)}{P(s_j | \bar{w})} \quad (3)$$

Using Equation 3, we rewrite the win ratio with respect to the strokes as shown below:

$$\frac{P(w | r)}{P(\bar{w} | r)} = \underbrace{\frac{P(w)}{P(\bar{w})}}_{\text{prior}} \cdot \prod_{j=1}^{|r|} \underbrace{\frac{P(s_j | w)}{P(s_j | \bar{w})}}_{\text{likelihood}} \quad (4)$$

We refer $\frac{P(w)}{P(\bar{w})}$ as the *prior* and $\frac{P(s_j | w)}{P(s_j | \bar{w})}$ as the *likelihood*.

In practice, computing the win ratio using Equation 2 would lead to numerical underflow due to the chain of products on the likelihoods. This issue was mitigated by applying natural logarithm on the win ratio to convert the products into sums. This led to the final formulation of the player-independent variant of the Naive Bayes rally outcome prediction model shown in Equation 5.

$$\ln \left(\frac{P(w | r)}{P(\bar{w} | r)} \right) = \underbrace{\ln \left(\frac{P(w)}{P(\bar{w})} \right)}_{\text{logprior}} + \sum_{j=1}^{|r|} \underbrace{\ln \left(\frac{P(s_j | w)}{P(s_j | \bar{w})} \right)}_{\text{loglikelihood}} \quad (5)$$

Using Equation 5, a rally r is predicted as the server wins if $\ln \left(\frac{P(w | r)}{P(\bar{w} | r)} \right) > 0$ or as the receiver wins if otherwise.

We next describe how to learn the model parameters *logprior* and *loglikelihood* from the dataset.

4.1.1 Logprior. Given a dataset $D = \{(r_i, W_i)\}$ which contains $|D|$ number of rallies r_i and outcomes $W_i \in \{w, \bar{w}\}$, calculating the model parameters is straightforward. Let N_w and $N_{\bar{w}}$ be the number of rallies won by the server and receiver respectively. The logprior can be calculated as follows:

$$\ln \left(\frac{P(w)}{P(\bar{w})} \right) = \ln \left(\frac{N_w}{N_{\bar{w}}} \right) \quad (6)$$

4.1.2 Loglikelihood. Calculating the loglikelihood requires $P(s_j | w)$ and $P(s_j | \bar{w})$ to be first obtained. In practice, we set s_j to be a stroke-player pair to additionally indicate who performed the stroke apart from the stroke itself. Specifically, s_j is an element of V where $V = S \times Q$ is a set of all possible stroke-player pairs. S is a set of stroke types in Table 2 and Q is the set of players $\{\text{server}, \text{receiver}\}$. We compute the loglikelihood for each stroke-player pair by tallying the number of stroke-player pairs observed in the dataset and accumulating the number of stroke-player pairs that contributes to server and receiver wins respectively.

$$\ln \left(\frac{P(s_j | w)}{P(s_j | \bar{w})} \right) = \ln \left(\frac{N_{ws_j}}{N_w} \cdot \frac{N_{\bar{w}}}{N_{\bar{w}s_j}} \right) \quad (7)$$

In Equation 7, N_{ws_j} and $N_{\bar{w}s_j}$ are the numbers of s_j observed in rallies won by the server and receiver respectively. N_w and $N_{\bar{w}}$ are the total numbers of stroke-pairs observed in rallies won by the server and receiver respectively.

4.2 Player-dependent Variant

The player-dependent variant can be easily extended from the player-independent variant by incorporating x and $\bar{x}$, the identities

Table 5: Details of matches between Kento MOMOTA and Anthony Sinisuka GINTING in our dataset.

No	Tournament	Results	Winner	Total Rallies Won		Total Rallies	Total Strokes
				MOMOTA	GINTING		
M1	2018 Blibli Indonesia Open	21-14, 21-15	MOMOTA	42	29	71	952
M2	2018 Victor China Open	21-23, 19-21	GINTING	40	44	84	1050
M3	2019 Daihatsu Indonesia Masters	21-9, 21-10	MOMOTA	42	19	61	733
M4	2019 Singapore Open	10-21, 21-19, 21-13	MOMOTA	52	53	105	1182
M5	2019 Total BWF Sudirman Cup	21-17, 21-19	MOMOTA	42	36	78	992
M6	2019 Daihatsu Yonex Japan Open	21-13, 20-22, 21-15	MOMOTA	62	50	112	1348
M7	2019 Victor China Open	19-21, 21-17, 21-19	MOMOTA	61	57	118	1392
M8	2019 Yonex French Open	10-21, 19-21	GINTING	29	42	71	753
M9	2019 HSBC BWF World Tour Finals	17-21, 21-17, 21-14	MOMOTA	59	52	111	1269
Total:				429	382	811	9671

of the server and the receiver respectively, into the Equation 5. This leads to the player-dependent model shown in Equation 8.

$$\ln \left(\frac{P_{x,\bar{x}}(w | r)}{P_{x,\bar{x}}(\bar{w} | r)} \right) = \underbrace{\ln \left(\frac{P_{x,\bar{x}}(w)}{P_{x,\bar{x}}(\bar{w})} \right)}_{\text{logprior}} + \sum_{j=1}^{|r|} \underbrace{\ln \left(\frac{P_{x,\bar{x}}(s_j | w)}{P_{x,\bar{x}}(s_j | \bar{w})} \right)}_{\text{loglikelihood}} \quad (8)$$

The major difference between the two variants lies in the manner in which the win frequencies are tallied. If the dataset consists of matches between various pairs of players, the player-independent variant, which is unconcerned about the actual players' identities, maintains only one table to tally win frequencies of servers and receivers. Conversely, the player-dependent variant needs to maintain one frequency table for each pair of players.

5 EXPERIMENTS, RESULTS, AND ANALYSIS

In this section, we evaluate the performance of our Naive Bayes model in rally outcome prediction for both player-independent and player-dependent variants. We seek to answer the following questions: **Question 1:** Does training a Naive Bayes model using rally stroke data helps to improve rally outcome prediction performance? **Question 2:** Does the player-dependent variant outperforms player-independent variant in rally outcome prediction?

5.1 Experimental Setup

Our experiments were implemented using Python 3.7.9 with Pympling⁶ and Numpy⁷ Python packages installed.

We preprocessed our dataset to remove rallies with erroneous events and with less than 3 strokes. We also converted the rally outcomes to either server wins or receiver wins. In particular, for rallies that ends with "miss", "lucky", and "misjudge", the winner is the player that performs the last stroke. For rallies that ends with "fault" or "out", the winner is the player that *does not* perform the last stroke. We did not differentiate forehand and backhand strokes in our experiments.

We used leave-one-match-out cross validation (LOMOCV) to split the dataset. This results in 8 matches in the training set and 1 match in the test set. We chose LOMOCV over the conventional

leave-one-out cross validation (LOOCV) as it reflects how human performs prediction in a match-by-match basis.

To gauge the effectiveness of our models in rally outcome prediction, we compute the accuracy and Brier Score (BS) as our performance metrics. Let $I_{\{A\}}$ be the indicator function of expression A such that $I_{\{A\}} = 1$ if A is true and $I_{\{A\}} = 0$ otherwise. The equation of accuracy and Brier Score are given in Equation 9 and 10 respectively as:

$$\text{Accuracy} = \frac{100}{N} \sum_i I_{\{\hat{y}^i = y^i\}} \quad (9)$$

$$\text{BS} = \frac{1}{N} \sum_i (P_{win}^i - p^i)^2 \quad (10)$$

In the equations, N is the number of rallies in the test set. $\hat{y}^i$ and y^i are the predicted and actual *converted* outcome (i.e. either the server wins or the receiver wins) of the i -th rally respectively. p^i is either 1 (if player x wins) or 0 (if player $\bar{x}$ wins) for the i -th rally. P_{win}^i is the predicted probability of player x winning the i -th rally. For our models, we apply the sigmoid function to compute P_{win} as follows:

$$P_{win} = \text{sigmoid} \left(\ln \left(\frac{P_{x,\bar{x}}(w | r)}{P_{x,\bar{x}}(\bar{w} | r)} \right) \right) \quad (11)$$

In every experiment described below, we use "NB-Independent" and "NB-Dependent" to refer to our player-independent and player-dependent models respectively to simplify the discussion.

5.2 Experiment 1: Effects of Rally Strokes Data on Rally Outcome Prediction Performance

This experiment aims to determine to effects of rally strokes data on the performance of rally outcome prediction. For this experiment, we conceptualized a fictitious method named "Oracle", which knows the winner of the match in question but not the strokes. The Oracle would naively predicts every rally in that match to be won by the winner. On the contrary, our models were trained with rally strokes data from other matches, but were not given the winner of the match in question. We compare the prediction performance between the Oracle and our models.

Table 6 depicts the prediction performance of Oracle and our models. The boldface value represents the best results on a given performance metric for a given match. Our models outperform

⁶<https://pypi.org/project/pympl-ing/>

⁷<https://numpy.org/>

Oracle in all matches except for match M3, despite Oracle knowing the winner of the match. These results suggest that rally strokes data and win frequencies help to improve the performance of rally outcome prediction. The Oracle performs well in match M3 because it is an outlier match in which MOMOTA wins GINTING convincingly. Our models' accuracies are inferior to Oracle in match M3 as they were trained on matches where the competitions between the players are stiff.

5.3 Experiment 2: Effects of Player-dependency on Rally Outcome Prediction Performance

Next, we compare NB-Independent and NB-Dependent to study the effects of player dependency on the performance of rally outcome prediction. From the overall results shown in Table 6, there is no strong indication that player dependency improves the performance of rally outcome prediction. In most matches, NB-Independent recorded higher accuracy compared to NB-Dependent, but were inferior to NB-Dependent in Brier Score. These results suggest that while NB-Independent is more accurate, it is also more uncertain in its prediction than NB-Dependent. Given the marginal difference in both accuracy and brier score between these two models as a whole, there is insufficient grounds from which to conclude that player-dependent model improves rally outcome prediction performance over player-independent model, contrary to our initial hypothesis.

To further understand the differences between our models, we compare the log likelihoods of both models on match M2 in which their accuracies are identical and on match M4 in which their accuracies vary the most. Table 7 and Table 8 compares the winning rates and log likelihoods for each stroke-player pair of both models for match M2 and M4 respectively. In both tables, the stroke-player pairs whose log likelihoods on both models are of different signs are highlighted. These are the strokes in which both models predict the winner differently. We observed that not only are there more stroke-pair in M4 than in M2 with differing log likelihood between NB-Independent and NB-Dependent, but the difference between the log likelihoods are also higher on average. This explains the disparity in accuracy in M4 between our models.

6 CONCLUSIONS

In this paper, we introduced a new badminton dataset for player-specific match analysis and prediction. We described how the match videos were selected when official match videos may be unavailable, and demonstrated how the videos were annotated with limited resources. We also proposed player-dependent and player-independent variants of Naive Bayes methods for rally outcome prediction and analyzed the performance of both baselines. In the future, we aim to expand our dataset over time by using cost-effective methodologies or automated approaches to collect and annotate future match videos. We hope our dataset will become a universal benchmark for comparing match analysis and prediction models, and help to expand the scientific understanding of badminton performance analysis.

ACKNOWLEDGMENTS

This research is supported by the Ministry of Education (MOE), Malaysia under the Fundamental Research Grant Scheme (FRGS) with Reference Code FRGS/1/2019/ICT02/MMU/03/6.

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Table 6: Results of Leave-one-match-out Cross Validation

Method	Metric	Matches								
		M1	M2	M3	M4	M5	M6	M7	M8	M9
Oracle	Accuracy (%)	59.15	52.38	68.85	50.48	53.85	55.36	51.69	59.15	53.15
NB-Independent		55.88	64.29	59.32	61.90	64.47	56.25	63.24	67.60	59.09
NB-Dependent		60.29	64.29	55.93	53.33	61.84	58.04	58.97	61.97	59.09
NB-Independent	Brier Score	0.259	0.301	0.281	0.290	0.303	0.288	0.295	0.302	0.278
NB-Dependent		0.278	0.297	0.273	0.273	0.292	0.288	0.287	0.291	0.279

Table 7: Match M2 - Comparison between player-independent and player-dependent models

Stroke	Player	NB-Independent					NB-Dependent				
		$x = \text{Server}$		$\bar{x} = \text{Receiver}$		Log Likelihood	$x = \text{GINTING}$		$\bar{x} = \text{MOMOTA}$		Log Likelihood
		Frequency	Probability	Frequency	Probability		Frequency	Probability	Frequency	Probability	
Block	x	294	0.0775	293	0.0750	0.015	305	0.0808	311	0.0793	-0.008
	$\bar{x}$	282	0.0744	329	0.0842	-0.054	264	0.0699	318	0.0810	-0.081
Clear	x	159	0.0419	188	0.0481	-0.060	148	0.0392	228	0.0334	-0.188
	$\bar{x}$	165	0.0435	120	0.0307	0.151	125	0.0331	131	0.0127	-0.020
Drive	x	40	0.0105	46	0.0118	-0.048	40	0.0106	50	0.0127	-0.097
	$\bar{x}$	47	0.0124	40	0.0102	0.083	43	0.0114	40	0.0102	0.031
Dropshot	x	221	0.0583	188	0.0481	0.083	160	0.0424	173	0.0441	-0.034
	$\bar{x}$	152	0.0401	164	0.0420	-0.020	167	0.0442	225	0.0573	-0.129
Lift	x	378	0.0998	390	0.0998	0.000	435	0.1152	499	0.1272	-0.060
	$\bar{x}$	580	0.1530	566	0.1448	0.024	471	0.1247	509	0.1297	-0.034
Net Kill	x	33	0.0087	21	0.0054	0.209	30	0.0079	21	0.0054	0.155
	$\bar{x}$	14	0.0037	30	0.0077	-0.318	14	0.0037	33	0.0084	-0.372
Net Shot	x	386	0.1018	465	0.1190	-0.068	417	0.1104	488	0.1244	-0.068
	$\bar{x}$	487	0.1284	373	0.0954	0.129	372	0.0985	434	0.1106	-0.067
Smash	x	327	0.0862	271	0.0693	0.095	277	0.0734	218	0.0556	0.104
	$\bar{x}$	211	0.0556	294	0.0752	-0.131	264	0.0699	344	0.0877	-0.115
Logprior = 0.1563						Logprior = -0.1541					

Table 8: Match M4 - Comparison between player-independent and player-dependent models

Stroke	Player	NB-Independent					NB-Dependent				
		$x = \text{Server}$		$\bar{x} = \text{Receiver}$		Log Likelihood	$x = \text{GINTING}$		$\bar{x} = \text{MOMOTA}$		Log Likelihood
		Frequency	Probability	Frequency	Probability		Frequency	Probability	Frequency	Probability	
Block	x	298	0.0786	300	0.0768	0.010	328	0.0869	318	0.0810	0.013
	$\bar{x}$	303	0.0799	329	0.0842	-0.023	285	0.0755	299	0.0762	-0.021
Clear	x	147	0.0388	169	0.0432	-0.047	144	0.0381	219	0.0558	-0.182
	$\bar{x}$	162	0.0427	116	0.0297	0.158	112	0.0297	119	0.0303	-0.026
Drive	x	44	0.0116	47	0.0120	-0.016	39	0.0103	49	0.0125	-0.099
	$\bar{x}$	41	0.0108	41	0.0105	0.013	39	0.0103	46	0.0117	-0.072
Dropshot	x	205	0.0541	195	0.0499	0.035	144	0.0381	176	0.0449	-0.087
	$\bar{x}$	158	0.0417	147	0.0376	0.044	177	0.0469	208	0.0530	-0.070
Lift	x	383	0.1010	387	0.0990	0.009	422	0.1118	472	0.1203	-0.049
	$\bar{x}$	571	0.1506	537	0.1374	0.040	486	0.1287	498	0.1269	-0.011
Net Kill	x	42	0.0111	18	0.0046	0.381	36	0.0095	18	0.0046	0.301
	$\bar{x}$	16	0.0042	28	0.0072	-0.230	16	0.0042	34	0.0087	-0.327
Net Shot	x	396	0.1044	456	0.1167	-0.048	433	0.1147	462	0.1177	-0.028
	$\bar{x}$	485	0.1279	346	0.0885	0.160	369	0.0977	419	0.1068	-0.055
Smash	x	330	0.0870	258	0.0660	0.120	290	0.0768	206	0.0525	0.149
	$\bar{x}$	212	0.0559	296	0.0757	-0.132	264	0.0699	336	0.0856	-0.105
Logprior = 0.1524						Logprior = -0.1408					

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